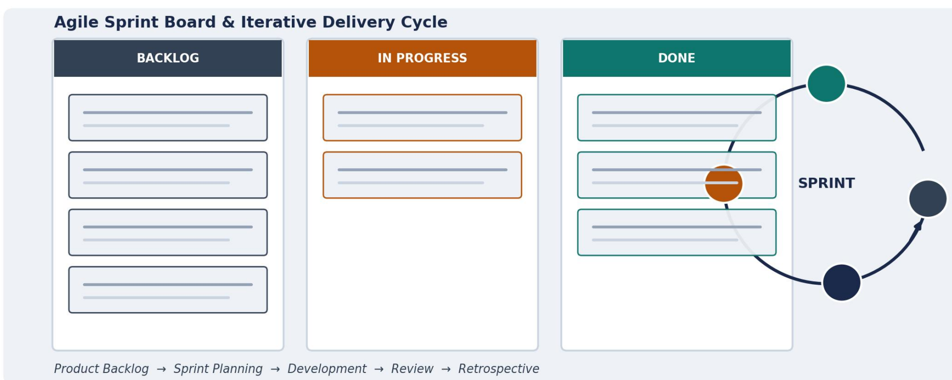
SOFTWARE ENGINEERING

AGILE SPRINT PLANNING & USER STORY WORKSHOP

C01 Assessment- 3

AI-Powered Smart Textile Manufacturing Quality Inspection System



Prepared By:	**Saruhaasini A**
Reg no.	**192511400**
Course Code:	**CSA1017**
Course Name:	**Software Engineering**

1. Project Vision and Stakeholders

1.1 Project Vision

Vision Statement

• To empower textile manufacturers with an AI-driven quality inspection platform that detects, classifies, and reports fabric defects in real time — transforming quality assurance from a slow, manual checkpoint into a continuous, intelligent, data-driven process that reduces waste, improves consistency, and strengthens customer trust.

1.2 Project Objectives

The Agile delivery plan is built around the following objectives, carried forward from the initial problem analysis and translated into a backlog of deliverable increments:

1. Detect textile defects automatically using AI-based computer vision.
2. Classify fabric quality and defect types with minimal human intervention.
3. Continuously monitor production-line quality in real time.
4. Reduce the number of defective products reaching packaging and dispatch.
5. Automatically generate quality-inspection reports and analytics dashboards.
6. Improve overall manufacturing efficiency and reduce inspection time.
7. Support predictive quality control through historical trend analysis.

1.3 Key Stakeholders

Effective sprint planning depends on a clear understanding of who defines value, who builds the product, and who will use or be affected by each increment. The stakeholders below participate in backlog refinement, sprint reviews, and retrospectives throughout the project.

Stakeholder	Role & Interest in the Sprint
Product Owner (Quality Assurance Head)	Owns and prioritizes the product backlog; represents business and customer interests; accepts/rejects completed increments.
Scrum Master	Facilitates sprint ceremonies, removes impediments, coaches the team on Agile practices, protects the team from scope creep.
Development Team (AI/ML	Designs, builds, tests, and delivers working increments of

Stakeholder	Role & Interest in the Sprint
Engineers, Software Developers, QA Testers)	the system each sprint; estimates story points collaboratively.
Textile Plant Owner / Top Management	Sponsors the project, sets strategic quality goals, and reviews sprint outcomes against business KPIs.
Production / Plant Manager	Provides shop-floor requirements, validates that increments fit real production workflows, attends sprint reviews.
QC Inspectors	Act as end users and domain experts; validate AI-flagged defects during demos; provide feedback each sprint.
Machine Operators	Represent front-line users of alerts and notifications; provide usability feedback on shop-floor interfaces.
IT / Infrastructure Team	Supports camera, network, and cloud infrastructure; ensures environments are ready for each sprint's deployment.
Customers / Buyers (Brands, Retailers)	Indirect stakeholders whose quality expectations shape acceptance criteria for reporting and traceability features.

2. Epics and User Stories

The product backlog is organized into six epics that map directly to the functional modules of the proposed system. Each epic is broken down into independent, testable user stories using the standard Agile format.

2.1 Epic Overview

Epic ID	Epic Name	Description
EPIC-1	Image Acquisition & Edge Preprocessing	Capture and prepare fabric images from production-line cameras for AI analysis.
EPIC-2	AI-Powered Defect Detection	Apply computer-vision models to automatically identify visual defects in real time.
EPIC-3	Defect Classification & Severity Scoring	Categorize detected defects and assign severity to support accept/reject decisions.
EPIC-4	Real-Time Quality Monitoring & Alerts	Provide live dashboards and threshold-based alerts to operators and managers.
EPIC-5	Reporting & Analytics	Generate automated reports and predictive insights from accumulated quality data.
EPIC-6	System Administration, Security & Integration	Manage access control, data security, and integration with ERP/MES systems.

2.2 User Stories by Epic

EPIC-1 — Image Acquisition & Edge Preprocessing

US-01
As a **machine operator**,
I want **the system to automatically capture continuous images of fabric as it moves through the production line**,
so that *no manual triggering is required to begin inspection*.

US-02
As a **IT engineer**,
I want **captured images to be pre-processed (noise reduction and normalization) at the edge**,
so that *only clean, relevant data is sent to the cloud for analysis*.

US-03
As a **system administrator**,
I want **to configure camera and lighting settings per production line**,
so that *image quality remains consistent across different fabric types*.

EPIC-2 — AI-Powered Defect Detection

US-04
As a **QC inspector**,
I want **the AI system to automatically detect defects such as holes, stains, and weaving faults in real time**,
so that *I do not have to visually scan every meter of fabric manually*.

US-05
As a **QC inspector**,
I want **the system to highlight the exact location of a detected defect on the fabric image**,
so that *I can quickly locate and verify it physically*.

US-06
As a **data scientist**,
I want **the ability to retrain the defect-detection model with newly labelled images**,
so that *detection accuracy improves as new defect types are encountered*.

EPIC-3 — Defect Classification & Severity Scoring

US-07
As a **QC inspector**,
I want **each detected defect to be automatically classified by type**,
so that *I can prioritize corrective actions appropriately*.

US-08
As a **production manager**,
I want **each defect to be assigned a severity score**,
so that *I can decide whether fabric should be accepted, reworked, or rejected*.

US-09
As a **QC inspector**,
I want **to manually confirm or override an AI classification**,
so that *edge cases are handled correctly and the system remains trustworthy*.

EPIC-4 — Real-Time Quality Monitoring & Alerts

US-10
As a **production manager**,
I want **a live dashboard showing current defect rates per line**,
so that *I can monitor quality without visiting the shop floor*.

US-11
As a **machine operator**,
I want **to receive an immediate alert when defect rates exceed a set threshold**,
so that *I can pause or adjust the machine before more defective fabric is produced*.

US-12
As a **plant manager**,
I want **to view historical quality trends by shift and product batch**,
so that *I can identify recurring quality issues*.

EPIC-5 — Reporting & Analytics

US-13
As a **quality manager**,
I want **the system to automatically generate daily and shift-wise quality reports**,
so that *I can review performance without manual compilation*.

US-14
As a **top management user**,
I want **to export quality reports in PDF/Excel format**,
so that *I can share them with customers and auditors*.

US-15

As a **data analyst**,
I want **predictive analytics on emerging defect trends**,
so that *I can anticipate quality issues before they escalate*.

EPIC-6 — System Administration, Security & Integration

US-16
As a **system administrator**,
I want **role-based access control across the platform**,
so that *only authorized users can view or modify sensitive quality data*.

US-17
As a **IT engineer**,
I want **the quality system to integrate with the existing ERP/MES system via APIs**,
so that *quality data is synchronized with production records automatically*.

US-18
As a **system administrator**,
I want **all data to be encrypted in transit and at rest**,
so that *production and quality data remain secure and compliant*.

3. Acceptance Criteria

Acceptance criteria are defined using the Given–When–Then format for each user story selected into Sprint 1 (see Section 5). This ensures every story has a clear, testable definition of done before development begins.

US-01 — Automatic Fabric Image Capture

Scenario 1: *Continuous capture during production*
Given the production line is running and the camera is powered on and connected
When fabric passes through the inspection zone
Then the system automatically captures images at the configured frame rate without manual triggering.

Scenario 2: *Camera disconnect handling*
Given the system is actively capturing images
When the camera connection is lost
Then the system logs the disconnection and raises an alert to the maintenance team within 30 seconds.

US-02 — Edge Image Pre-processing

Scenario 1: *Noise reduction and normalization*
Given a raw image has been captured from the production line
When the image reaches the edge pre-processing module
Then the system applies noise reduction and normalization and forwards only the cleaned image to the cloud.

Scenario 2: *Bandwidth-constrained network*
Given network bandwidth to the cloud is limited
When pre-processed images are queued for transmission
Then the system compresses images without losing defect-relevant detail before upload.

US-03 — Camera and Lighting Configuration

Scenario 1: *Per-line configuration*
Given an administrator is logged into the system configuration panel
When they select a production line and adjust camera resolution or lighting intensity
Then the new settings are applied to that line's camera immediately and saved for future sessions.

Scenario 2: *Invalid configuration rejected*
Given an administrator attempts to set a value outside the supported range
When they submit the configuration form
Then the system rejects the change and displays a clear validation message.

US-04 — Real-Time AI Defect Detection

Scenario 1: *Defect correctly detected*
Given a pre-processed fabric image containing a known defect (hole, stain, or weave fault) is submitted to the AI engine
When the model completes inference
Then the system flags the image as defective with at least 90% confidence and records the defect type.

Scenario 2: *Defect-free fabric passes*
Given a pre-processed image of defect-free fabric is submitted to the AI engine
When the model completes inference
Then the system marks the image as passed with no false defect flag.

Scenario 3: *Inference within latency budget*
Given the AI defect detection engine is running under normal production load
When an image is submitted for analysis
Then the system returns a detection result within 1 second to keep pace with the production line.

US-05 — Defect Location Highlighting

Scenario 1: *Bounding box overlay shown*
Given the AI engine has flagged an image as defective
When a QC inspector opens the image in the inspection UI
Then the system displays a bounding box or marker precisely over the detected defect location.

Scenario 2: *Multiple defects on one frame*
Given an image contains more than one defect
When the QC inspector views the flagged image
Then the system displays a separate marker for each detected defect, labelled individually.

4. Product Backlog Prioritization

4.1 Prioritization Approach

The backlog is prioritized using a combination of the MoSCoW technique (Must Have / Should Have / Could Have / Won't Have) and a Business Value vs. Effort matrix. MoSCoW establishes release-level priority aligned with the project's core objectives, while the value/effort matrix helps the Product Owner and team sequence work within each priority tier — favouring high-value, lower-effort "quick wins" early, while scheduling high-effort/high-value "major" items across multiple sprints.

4.2 Full Product Backlog

ID	Epic	User Story (Summary)	Priority	Value (1–10)	Story Points	Dependency
US-01	EPIC-1	Automatic fabric image capture	Must Have			None
US-02	EPIC-1	Edge image pre-processing	Must Have			US-01
US-03	EPIC-1	Camera/lighting configuration	Should Have			US-01
US-04	EPIC-2	Real-time AI defect detection	Must Have			US-02
US-05	EPIC-2	Defect location highlighting	Must Have			US-04
US-06	EPIC-2	Retrain detection model with new data	Should Have			US-04
US-07	EPIC-3	Automatic defect classification	Must Have			US-04
US-08	EPIC-3	Defect severity scoring	Must Have			US-07
US-09	EPIC-3	Manual confirm/override of AI classification	Should Have			US-07
US-10	EPIC-4	Live quality dashboard	Must Have			US-07
US-11	EPIC-4	Threshold-based real-time alerts	Must Have			US-10
US-12	EPIC-4	Historical quality trend view	Should Have			US-10
US-13	EPIC-5	Automated shift/daily quality reports	Should Have			US-10

ID	Epic	User Story (Summary)	Priority	Value (1–10)	Story Points	Dependency
US-14	EPIC-5	Export reports to PDF/Excel	Could Have			US-13
US-15	EPIC-5	Predictive defect-trend analytics	Could Have			US-13
US-16	EPIC-6	Role-based access control	Must Have			None
US-17	EPIC-6	ERP/MES API integration	Should Have			US-13
US-18	EPIC-6	Data encryption in transit and at rest	Must Have			US-16

4.3 Business Value vs. Effort Matrix

The chart below plots every backlog item by business value and estimated effort (story points). Items in the upper-right quadrant deliver both high value and require higher investment — these anchor the release roadmap. Items highlighted in amber were selected for Sprint 1, chosen for forming a coherent, demonstrable "walking skeleton" of the core detection pipeline.

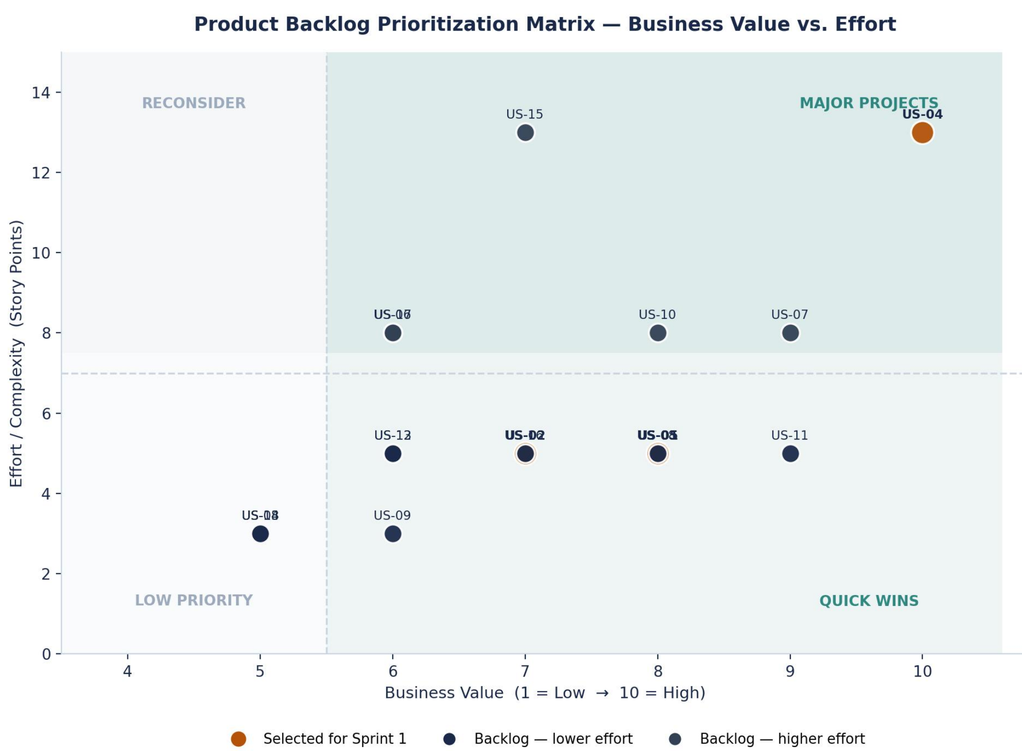


Figure 4.1 — Product backlog items plotted by business value and effort, with Sprint 1 selections highlighted.

5. Sprint Backlog Planning

5.1 Sprint 1 Selection Rationale

Sprint 1 focuses on delivering a coherent, end-to-end "walking skeleton" of the core detection pipeline rather than isolated features. The five selected stories — automatic image capture, edge pre-processing, camera/lighting configuration, real-time AI defect detection, and defect-location highlighting — together allow the team to demonstrate, at the end of the sprint, fabric moving through a live camera feed with AI-detected defects visibly flagged for a QC inspector. This vertical slice validates the riskiest technical assumptions (camera integration and model latency) early, in line with Agile risk-first sequencing.

5.2 Sprint Backlog: Selected User Stories

ID	User Story Summary	Story Points	Priority
US-01	Automatic fabric image capture		Must Have
US-02	Edge image pre-processing		Must Have
US-03	Camera/lighting configuration		Should Have
US-04	Real-time AI defect detection		Must Have
US-05	Defect location highlighting		Must Have

Total Sprint 1 Commitment: 31 Story Points (against an assumed team velocity of 30–32 SP per 2-week sprint).

5.3 Task Breakdown

Each selected user story is decomposed into actionable development tasks assigned during sprint planning, enabling daily progress tracking on the sprint board.

Story ID	User Story	Development Task
US-01	Automatic fabric image capture	Integrate industrial camera SDK with edge capture service
		Implement continuous frame-capture loop synced to line speed
		Add camera health-check and reconnection logic
		Unit test capture service under simulated line conditions
US-02	Edge image pre-processing	Implement noise-reduction filter on captured frames
		Implement image normalization (resolution, brightness,

Story ID	User Story	Development Task
		contrast)
		Build compression step for bandwidth-constrained upload
		Integration test: capture → pre-process → upload pipeline
US-03	Camera/lighting configuration	Design configuration schema per production line
		Build admin UI form for camera/lighting parameters
		Implement validation rules and error messaging
		Persist configuration and apply on camera restart
US-04	Real-time AI defect detection	Prepare and augment labelled training dataset (holes, stains, weave faults)
		Train and validate baseline CNN defect-detection model
		Deploy model as a low-latency inference microservice
		Benchmark inference latency against 1-second target
		Integration test with live pre-processed image stream
US-05	Defect location highlighting	Extract bounding-box coordinates from model output
		Build overlay rendering component for inspection UI
		Support multiple simultaneous defect markers per image
		Usability review with a QC inspector (stakeholder feedback)

6. Story Point Estimation

6.1 Estimation Technique

The team estimates effort using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21…). Each team member privately selects a card representing their effort estimate for a story; differing estimates are discussed openly to surface hidden assumptions or risks before converging on a consensus value. The non-linear spacing of the Fibonacci sequence reflects the natural increase in estimation uncertainty for larger, more complex stories.

6.2 Sprint 1 Story Point Estimates and Justification

Story ID	Story Points	Estimation Rationale
US-01		Requires industrial camera SDK integration and a continuous capture loop synchronized to variable line speed; moderate complexity with some hardware-interfacing uncertainty.
US-02		Involves multiple image-processing steps (noise reduction, normalization, compression) that must run reliably at line speed; similar complexity to US-01.
US-03		A relatively standard CRUD-style configuration feature (form, validation, persistence) with well-understood scope and low technical risk.
US-04		The highest-risk, highest-complexity story: involves dataset preparation, model training/validation, deployment as a low-latency service, and meeting a strict latency target — significant uncertainty warrants the largest estimate.
US-05		Depends on US-04's output format; overlay rendering itself is moderate complexity, but coordination with model output and multi-defect support adds some uncertainty.

6.3 Story Point Distribution

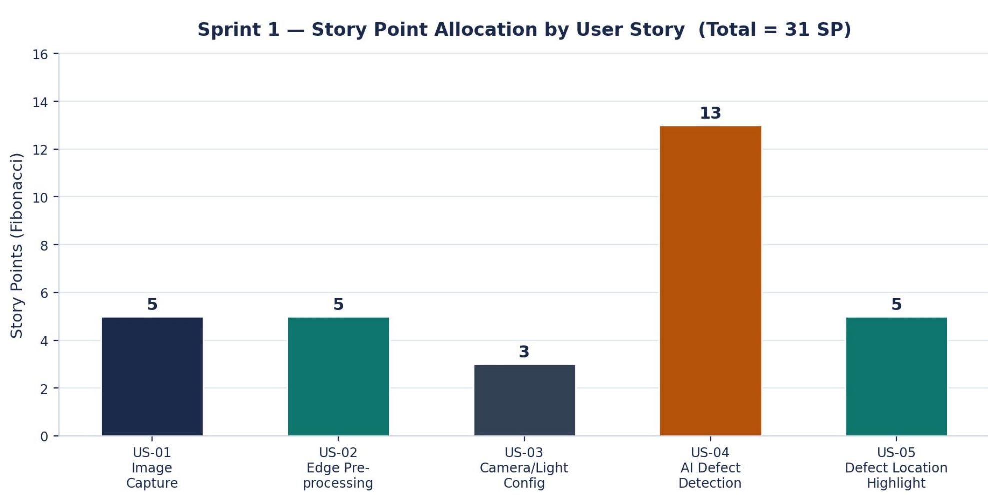


Figure 6.1 — Story point allocation across Sprint 1 user stories, totalling 31 story points.

6.4 Planned Sprint Burndown

Based on the 31-point commitment and a 10-working-day sprint, the team projects a burndown trajectory below. A short plateau is anticipated around days 2–4 while the AI model (US-04) is trained and validated — the highest-uncertainty task — followed by faster completion as remaining UI and integration tasks are finished.

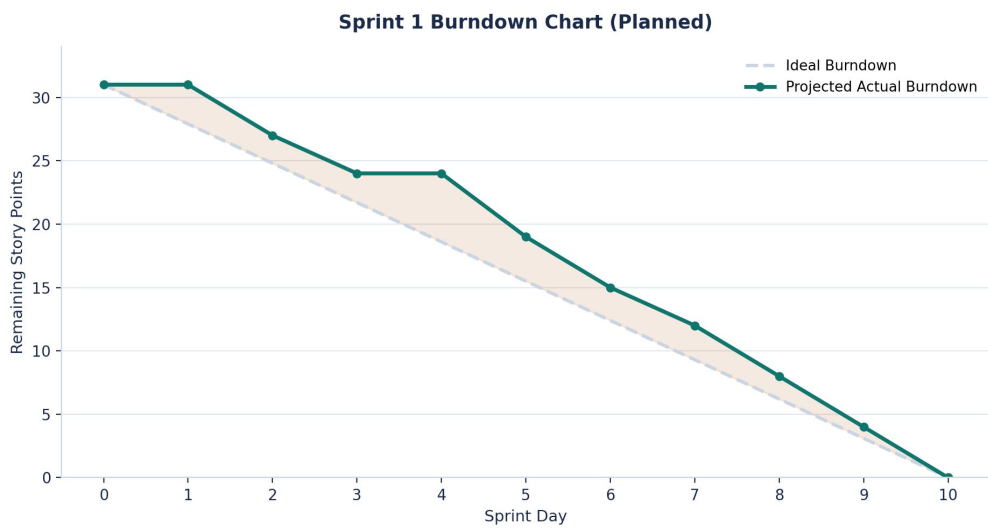


Figure 6.2 — Planned Sprint 1 burndown showing the ideal linear trajectory against a realistic projected path.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

Sprint 1 Goal

• Establish the foundational image-acquisition and AI-based defect-detection pipeline, enabling the system to automatically capture fabric images from a pilot production line, pre-process them at the edge, detect defects in real time using a baseline AI model, and visually flag defect locations for QC inspectors.

7.2 Expected Sprint Deliverables

- A working camera-integration module streaming live fabric images from at least one pilot production line.
- An operational edge pre-processing pipeline performing noise reduction, normalization, and compression.
- A configuration interface allowing administrators to adjust camera and lighting settings per line.
- A deployed AI defect-detection engine capable of identifying at least three defect types (holes, stains, weaving faults) with ≥90% confidence on test data.
- A basic inspection UI that overlays detected-defect locations on captured images for QC inspector review.
- A demo-ready increment presented at the Sprint Review to the Product Owner, QC lead, and Plant Manager, along with a retrospective capturing lessons learned for Sprint 2 planning.

7.3 Sprint Metadata

Attribute	Value
Sprint Duration	2 weeks (10 working days)
Sprint Number	Sprint 1 of the project roadmap
Team Composition	1 Product Owner, 1 Scrum Master, 2 AI/ML Engineers, 2 Software Developers, 1 QA Tester
Assumed Team Velocity	30–32 story points per sprint
Sprint Commitment	31 story points (5 user stories across EPIC-1 and EPIC-2)
Definition of Done	Code reviewed and merged; unit and integration tests passing; acceptance criteria verified; demoed in Sprint Review; documentation updated

Conclusion

This sprint planning document translates the AI-Powered Smart Textile Quality Inspection System's problem statement into an actionable Agile delivery plan. By organizing requirements into six epics and eighteen well-formed user stories, prioritizing the backlog through a combined MoSCoW and value/effort approach, and committing to a focused, risk-first Sprint 1, the team establishes a clear path toward incrementally delivering a working, demonstrable system. The chosen Sprint 1 scope — camera integration, edge pre-processing, and real-time AI defect detection — validates the project's most technically uncertain assumptions early, positioning subsequent sprints to build classification, monitoring, and reporting capabilities on a proven foundation.